École Nationale des Ponts et Chaussées

Département de première année

Bases scientifiques pour la transition énergétique 1

Séance 5 : Récapitulatif des points essentiels partie nucléaire

2026

Recap of what must be known for the exam

1. Nuclei

A nucleus, noted ${}^{A}_{Z}\mathrm{X}$ with X the symbol of the element, is characterized by its **number of charge** Z (number of protons) and its **number of mass** A (number of protons and neutrons). The number of charges determines how the electrons arrange around the nucleus. Thus, two nuclei with same Z but different A, called **isotopes**, have identical chemical properties, and differ only by their mass.

Because of the energy-mass equivalence of relativity ($E = mc^2$ for a particle of mass m at rest), any change of mass characterizes a change of energy. Therefore, the difference of mass between an atom and the same number of free protons and free neutrons characterizes the binding energy of an atom (the mass of the electrons is negligible):

$$B\left(Z,A\right) = Z \times M_{\mathrm{H}} + \left(A - Z\right) \times M_n - M_{\mathrm{atom}}\left(Z,A\right)$$

Binding energies are usually small compared to the total mass, so one often refers to the concept of **mass excess** instead of the total mass. The mass excess is the difference between the total mass and A times a reference mass (atomic mass unit u or 'dalton', corresponding to $1/12^{th}$ of the mass of ${}^{12}_{6}\mathrm{C}$:

$$\Delta\left(Z,A\right) = M_{\mathrm{atom}}\left(Z,A\right) - Au \text{ with } 1u = \frac{1}{12} M_{{}^{12}_{6}\mathrm{C}} = 1 \text{ dalton}$$

Since all nuclear reactions of interest for energy applications conserve the number of mass between the reactants and the products, the balance of energy of a nuclear reaction is the same when expressed in terms of mass or of mass excess:

$$Q_{\mathrm{reaction}} = \sum_{\mathrm{reactants}} M_i - \sum_{\mathrm{products}} M_i = \sum_{\mathrm{reactants}} \Delta_i - \sum_{\mathrm{products}} \Delta_i$$

2. Nuclear decay

Nuclear reactions can vary the composition of the nuclei in terms of protons and neutrons provided that the number of mass A is conserved. Accordingly, the binding energy per nucleon $b\left(Z,A\right) = \frac{B(Z,A)}{A}$ can be used to compare the level of energy of the nuclei. $b\left(Z,A\right)$ exhibits a '**valley of stability**' (see Fig. **??**) and ${}^{56}_{26}\mathrm{Fe}$ is the most tightly bound nucleus. Nuclei lighter than ${}^{56}_{26}\mathrm{Fe}$ can fusion to produce a heavier nucleus, whereas a nucleus heavier than ${}^{56}_{26}\mathrm{Fe}$ can

fission to produce lighter nuclei. The 'valley of stability' is quite narrow, only a few isotopes for each atom can exist.

Nuclei can reduce their binding energy by nuclear decays, i.e., nuclear reactions by which a nucleus can spontaneously emit a small particle to change Z and/or decrease A. **Nuclear decays** are random processes and the probability that a nucleus has not decayed decreases exponentially with time $\exp\left(-\frac{\ln(2)}{\tau_{1/2}}t\right)$. The characteristic time $\tau_{1/2}$ represents the **half-life time**. The most frequent nuclear decays are :

β^- **decay:** a neutron is converted into a proton

$$^{A}[\mathrm{Z}] \rightarrow {}^{A}[\mathrm{Z}+1] + e^- + \bar{\nu}_e$$

β^+ **decay:** a proton is converted into a neutron

$$^{A}[\mathrm{Z}] \rightarrow {}^{A}[\mathrm{Z}-1] + e^+ + \nu_e$$

electron capture: a proton combines with an electron to form a neutron

$$^{A}[\mathrm{Z}] + e^- \rightarrow {}^{A}[\mathrm{Z}-1] + \nu_e$$

α **emission:** emission of an $^{4}_{2}\mathrm{He}$

$$^{A}[\mathrm{Z}] \rightarrow {}^{A-4}[\mathrm{Z}-2] + {}^{4}_{2}\mathrm{He}$$

β^+ is less frequent than e^- (releases less energy) while connecting the same nuclei. Thus, one often refers to β^- decay as to β decay.

A few heavy nuclei also exhibit **spontaneous fission**, a process by which the nucleus breaks into two lighter nuclei of similar size. Spontaneous fission is a rare event compared to the other decays, but it produces free neutrons, which can serve to initiate nuclear chain reactions.

A **decay chain** is a sequence of decay reactions that follows a unstable nucleus to reach stability. There are only four primary decay chains that can be observed in nature starting from $^{232}_{90}\mathrm{Th}$, $^{237}_{93}\mathrm{Np}$, $^{238}_{92}\mathrm{U}$, and $^{235}_{92}\mathrm{U}$.

Nuclei resulting from a nuclear reaction can be found in excited states of energy, called nuclear isomers. Nuclear isomers are metastable and they ultimately relax to the non-excited state of the respective nuclei by emitting high energy photons (MeV) in the γ-ray domain. These γ **emissions** are particularly harmful for living organisms as they can penetrate deep into matter and ionize any atom.

3. nuclear reactions

A nuclear reaction is a process during which two nuclei interact to produce one or several new nuclei. Of particular interest are:

Neutron capture: a neutron is captured by a nucleus leading to a heavier excited nucleus ('*') that may further decay. If the decay is a gamma emission, the reaction reaction is called a **radiative capture**

$$^{A}[\mathrm{Z}] + {}^{1}_{0}\mathrm{n} \rightarrow {}^{A+1}[\mathrm{Z}]^*$$

Induced fission: after a neutron capture, the nucleus breaks into lighter nuclei.

$$^{A}[\mathrm{Z}] + {}^{1}_{0}\mathrm{n} \rightarrow {}^{a}[\mathrm{z}] + {}^{A-a-x}[\mathrm{Z}-\mathrm{z}] + (x+1)\,{}^{1}_{0}\mathrm{n}$$

Induced fission is possible only for a few heavy nuclei that are said 'fissile'. $^{235}_{92}$U is the only naturally occurring fissile nuclei. Some non-fissile nuclei are said 'fertile' because they can produce fissile nuclei after a radiative capture. Fission is a stochastic process, and there is a wide variety of possible fission products. Induced fission also produces a few neutrons of high kinetic energy (fast neutrons).

Fusion: Two light nuclei with enough kinetic energy can come close enough to form a single heavier nucleus. Fusion requires extreme temperatures and takes place in particular in stars. Fusion between deuterium and tritium is actively studied for energy applications:

$$^{2}_{1}\mathrm{H} + ^{3}_{1}\mathrm{H} \rightarrow ^{4}_{2}\mathrm{He} + ^{1}_{0}\mathrm{n}$$

Neutron scattering is the process by which a neutron interacts with a nucleus but is not captured. Neutron scattering is not a nuclear reaction but leads to an exchange of energy with the nucleus.

For a nucleus submitted to an incoming flux ϕ of particles (number of particles per unit area and time), the probability that a nuclear process occurs per unit time is proprotional to the flux: $\frac{dp_{\text{reaction}}}{dt} = \sigma\phi$. The proportionality factor σ has a unit of area and is called the cross-section.

4. fission reactors

The vast majority of nuclear reactors use the induced fission of $^{235}_{92}$U by thermal neutrons to produce heat which is then converted to electricity. Key physical principles are:

- This fission reaction produces 2.42 fast neutrons on average. So a chain reaction is possible if at least 1 one those 2.42 fast neutrons triggers another fission.
- The cross section of $^{235}_{92}$U induced fission is much smaller for fast neutrons than for thermal neutrons. So the fast neutrons produced by the fission must be slowed down to allow a chain reaction. This is done by using a 'moderator', i.e., a medium made of light nuclei with high scattering cross section (to easily transfer the kinetic energy of the fast neutrons to the medium) and low capture cross section (to avoid consuming the neutrons). Water is frequently used as a moderator.
- During the thermalization process, part of the neutrons are inevitably lost because of 1) radiative capture by $^{238}_{92}$U, by the moderator or by some fission products with high capture cross-section (e.g., $^{135}_{54}$Xe), 2) leakage at the boundary of the reactor.

The reproduction factor (number of fast neutrons produced on average for each thermal neutron capture) of natural uranium (0.72% of $^{235}_{92}$U) is only $\eta = 1.33$. Unless a very efficient moderator is used (e.g., heavy water), a nuclear reactor is unable to maintain a chain reaction with natural uranium. Therefore, most nuclear reactors use enriched uranium with 3.5% to 5% of $^{235}_{92}$U.